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QSCP-IP-ATU100 Rev 1	Issued	2008-02-15

Piping Inspection Plan - ATU-100

Scope and Objectives

This document defines the mechanical integrity strategy for the piping systems within ATU-100 at the Hamilton Plant. The primary objective is to manage localized corrosion risks identified in the regenerator overhead circuit, specifically addressing the vulnerabilities of carbon steel components in acid gas service. Douglas Ferreira, acting as the Inspection Supervisor and API 570 inspector, is responsible for the technical oversight and implementation of these monitoring requirements.

This revision supersedes QSCP-IP-ATU100 Rev 0 to incorporate findings from recent integrity failures. The scope includes the reassessment of 3-RV-101 and 2-RV-102, ensuring that all pressure-retaining components are monitored according to their specific degradation profiles. The Hamilton Inspection Department shall ensure all thickness data is recorded against the updated CML distribution to maintain compliance with site-specific remaining life calculations.

Technical Basis and Reference Analysis

The modifications to this plan are driven by QSCP-RCA-2007-002, which investigated the 2-RV-102 elbow leak. The analysis determined that localized corrosion from condensing acid gas caused the breach, exacerbated by a lack of monitoring points on fittings. Previously, QSCP-IP-ATU100 Rev 0 only required CMLs on straight pipe runs, which failed to capture the accelerated thinning occurring at directional changes.

In response to these findings, the failed elbow on 2-RV-102 has been replaced with A106 Gr.B material. The total CML count for the unit has been increased by 20 fittings, bringing the total to 138 monitoring locations. This expansion ensures that high-turbulence areas and condensation points are adequately represented in the corrosion rate data set.

Inspection Intervals and Circuit Classification

The regenerator overhead circuit, containing 3-RV-101 and 2-RV-102, is classified as Class 1 service due to the presence of sour gas. While standard Class 1 maximum intervals are 5 years, this circuit is now restricted to a 3-year thickness inspection interval. This conservative frequency is mandated to mitigate the risks of localized acid gas corrosion identified in QSCP-RCA-2007-002.

Remaining life for these lines is calculated using the larger of the long-term and short-term corrosion rates. The next inspection date must be scheduled at no more than half the remaining life, not to exceed the 3-year circuit limit. Any thickness reading recorded below the t-min of 3.56 mm for 3-RV-101 or 2.79 mm for 2-RV-102 must be immediately raised as an NCR.

Responsibilities and Execution

Douglas Ferreira is responsible for reviewing all NDE data and verifying that the short-term corrosion rates, calculated over a minimum 3-year span, accurately reflect current process conditions. The Hamilton Inspection Department must ensure that the 20 new fitting CMLs are clearly identified in the field to prevent the recurrence of spot-reading errors noted in previous assessments.

Field inspectors are required to report any external visual anomalies during the 5-year visual inspection cycle for Class 1 and Class 2 lines. All data must be integrated into the Hamilton Plant asset management system, ensuring that the transition from the requirements of QSCP-IP-ATU100 Rev 0 to this revised plan is fully documented and traceable for future audits.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2008-02-15	_____